FLYBOAT

Flyboat Electronic LLC — North American sales & engineering center

25D Series Metal Oxide Varistors

Epoxy-encapsulated zinc oxide varistors · Datasheet

The 25D series is a line of 25 mm round, epoxy-encapsulated zinc oxide varistors made by Guiyang High-Tech Yige Electronic Co., Ltd. for surge protective device (SPD) manufacturers and OEMs. Rated In 10 kA / Imax 20 kA (8/20 µs), the range covers varistor voltages from 100 V to 820 V in 22 model grades. The varistor discs are pressed, sintered, metallized, coated and tested in our own Guiyang plant — not bought in — which is how tolerances and surge ratings stay consistent from batch to batch. In North America, the series is supported by Flyboat Electronic LLC, the factory's sales and engineering center in Bellevue, WA.

Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- For top grades the melt-through point is confined within a 0.5 cm radius of the chip centre, converting a random failure location into a controlled one.
- If the chip does melt through, the flame jet is shielded by the copper electrode, which conducts the heat to the trip point and drives the thermal disconnector open.
- Exceptional damp-heat resistance: under a customer's severe boil test (30 min in boiling water, cool, measure, re-boil; 6 cycles required with no significant change in varistor voltage or leakage), Yige product has reached 8 cycles with small-current parameters essentially unchanged.
- Long, verifiable service life (Arrhenius law): verified minimum expected life of 110 years at 40 °C, 11 years at 65 °C, 0.7 years at 95 °C upper ambient temperature (115 °C / 1,000 h ageing basis).
- High TOV withstand: at a temporary overvoltage of 1.45 × Uc the chip stays thermally stable for more than 45 min (r.m.s.), with no damage.

Certification status

The 25D series is not UL recognized: the catalog ratings tables for this series print the UL / C-UL / TÜV columns but leave every cell blank. For series not yet UL recognized, we can pursue UL certification for your program.
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Model naming: 25D K 431 — 25 = nominal size (mm); K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead / structure form.

U1mA = varistor voltage at DC 1 mA; ΔU1mA = tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In (reference value–ratio); ETM = maximum energy absorption (2 ms).

25D standard ratings — full table

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
25DK101	100	±10	60	85	10	20	360-3.6	90
25DK121	120	±10	75	100	10	20	400-3.6	110
25DK151	150	±10	95	125	10	20	450-3.0	160
25DK181	180	±10	115	150	10	20	490-2.8	140
25DK201	200	±10	130	170	10	20	520-2.6	200
25DK221	220	±10	140	180	10	20	550-2.6	220
25DK241	240	±10	150	200	10	20	600-2.6	235
25DK271	270	±10	175	225	10	20	680-2.6	270
25DK301	300	±10	195	250	10	20	770-2.6	300
25DK331	330	±10	210	270	10	20	830-2.55	330
25DK361	360	±10	230	300	10	20	910-2.55	370
25DK391	390	±10	250	320	10	20	1000-2.55	410
25DK431	430	±10	275	350	10	20	1050-2.45	460
25DK471	470	±10	300	385	10	20	1100-2.4	500
25DK511	510	±10	320	410	10	20	1150-2.35	530
25DK561	560	±10	350	450	10	20	1250-2.35	580
25DK621	620	±10	385	505	10	20	1550-2.35	630
25DK681	680	±10	420	560	10	20	1650-2.35	640
25DK711	710	±10	440	585	10	20	1700-2.35	650
25DK751	750	±10	460	615	10	20	1700-2.3	670
25DK781	780	±10	485	640	10	20	1750-2.3	690
25DK821	820	±10	510	670	10	20	1850-2.3	720

25D low-residual-voltage ratings

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
25DK431	430	±10	275	350	10	20	990-2.2	460
25DK471	470	±10	300	385	10	20	1070-2.2	500
25DK511	510	±10	320	410	10	20	1150-2.2	530
25DK561	560	±10	350	450	10	20	1250-2.15	580
25DK621	620	±10	385	505	10	20	1350-2.1	630
25DK681	680	±10	420	560	10	20	1520-2.1	640

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
25DK711	710	±10	440	585	10	20	1600-2.1	650
25DK751	750	±10	460	615	10	20	1700-2.1	670
25DK781	780	±10	485	640	10	20	1760-2.1	690
25DK821	820	±10	510	670	10	20	1800-2.08	720

Ratings as published at www.flyboatmov.com/25d-series.html.

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